Original research

Deep learning on pre-procedural computed tomography and clinical data predicts outcome following stroke thrombectomy

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ABSTRACT

Background Deep learning using clinical and imaging data may improve pre-treatment prognostication in ischemic stroke patients undergoing endovascular thrombectomy (EVT).

Methods Deep learning models were trained and tested on baseline clinical and imaging (CT head and CT angiography) data to predict 3-month functional outcomes in stroke patients who underwent EVT. Classical machine learning models (logistic regression and random forest classifiers) were constructed to compare their performance with the deep learning models. An external validation dataset was used to validate the models. The MR PREDICTS prognostic tool was tested on the external validation set, and its performance was compared with the deep learning and classical machine learning models.

Results A total of 975 patients (550 men; mean±SD age 67.5±15.1 years) were studied with 778 patients in the model development cohort and 197 in the external validation cohort. The deep learning model trained on baseline CT and clinical data, and the logistic regression model (clinical data alone) demonstrated the strongest discriminative abilities for 3-month functional outcome and were comparable (AUC 0.811 vs 0.817, Q=0.82). Both models exhibited superior prognostic performance than the other deep learning (CT head alone, CT head, and CT angiography) and MR PREDICTS models (all Q<0.05).

Conclusions The discriminative performance of deep learning for predicting functional independence was comparable to logistic regression. Future studies should focus on whether incorporating procedural and post-procedural data significantly improves model performance.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Prognostic tools developed in endovascular thrombectomy patients using statistical modeling have reported modest discriminative ability, which has limited their clinical utility.

WHAT THIS STUDY ADDS

⇒ This study showed that the performance of deep learning using raw inputs from CT head, CT angiography, and clinical data was similar to the performance of classical logistic regression prognostic models.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ The performance of our deep learning model could be improved by re-training the model with larger datasets, or by incorporating peri- or post-procedural parameters into the training datasets.

INTRODUCTION

Endovascular thrombectomy (EVT) is now supported by level I evidence for most patients with large vessel occlusion (LVO) ischemic stroke, including those with large ischemic cores,[1] and basilar artery occlusions.[2] However, between one-third and two-thirds of anterior circulation LVO patients treated with EVT are left non-ambulatory or dead at 3 months.[3 4] A prognostic tool that can be used when considering EVT could provide physicians, patients, and family members with reliable outcome expectations. Prognostic tools developed in EVT patients using statistical modeling such as the ordinal logistic regression model, MR PREDICTS, have reported modest discriminative ability, which has limited their clinical utility.[5] Consequently, there is growing interest in using novel modeling techniques, such as deep learning, to develop better performing predictive models.[6]

Statistical modeling and classical machine learning are related techniques that use algorithms to create regression or classification models.[7] Examples include linear regression, logistic regression, decision trees, support vector machines, and random forests.[8] Logistic regression, widely used to develop clinical prognostic tools, takes a weighted linear combination of inputs with a bias term, and transforms it with a sigmoid function to produce an output between 0 and 1.[9] The weights and bias are found with an optimization algorithm that minimizes a cost function across a collection of training samples.[9]

Deep learning is a type of artificial intelligence that uses multilayered artificial neural networks to learn progressively higher-level features from raw input data, such as a two- or three-dimensional medical images.[10] Logistic regression assumes a linear relationship between the log odds of an outcome and the independent variables,[9] whereas deep learning models can use nonlinear relationships to predict outcomes.[8] As these models can capture nonlinearity, it is a common practice to feed raw input to the model without feature engineering. In medical image processing, lower model layers identify 'low-level' features such as edges, corners, textures, or colors in an image, while higher model layers identify 'high-level' clinically relevant concepts, such as infarct volume[11] or intracranial hemorrhage.[12]

Deep learning has the potential advantage over statistical and classical machine learning models of being able to automatically analyze raw imaging data without the need for human interpretation, and has shown promise in predicting outcomes following EVT using pre-procedural CT angiography (CTA).[13] We aimed to develop a deep learning model using pre-procedural non-contrast CT (NCCT), CTA, and clinical data to predict functional independence following EVT. The deep learning models were compared with the best performing classical machine learning model (logistic regression or random forest classifier) and MR PREDICTS.

METHODS

Study design

The study is reported according to the transparent reporting of a multivariable prediction model for individual prognosis or diagnosis statement.[14] The code for the final models are published open source on https://github.com/jdddog/deep-mt and https://github.com/jdddog/deep-skull. The pre-trained model weights and data that support the findings of this study are available from the corresponding author on reasonable request. This study was approved by the regional ethics committee (NZ/1/E82215) and informed consent was not required. Consecutive EVT patients from two comprehensive stroke centers were included retrospectively. Patients without an NCCT brain scan on the regional hospital Picture Archiving and Communication System (PACS) or without 3-month follow-up were excluded. Data from only the most recent EVT admission was used if a patient underwent EVT on more than one occasion during the study period. The National Institutes of Health Stroke Scale (NIHSS) score at baseline was assessed by the attending stroke teams. The Alberta Stroke Program Early CT Score (ASPECTS) was used to assess the size of baseline infarct. Successful recanalization was defined as a modified Thrombolysis in Cerebral Infarction (TICI) score of 2b to 3. Modified Rankin Scale (mRS) scores were determined 3 months after the stroke and functional independence was defined as an mRS of 0–2.

Data elements, clinical and image pre-processing

The features and pre-processing steps are described in detail in the online supplemental material. A total of 26 clinical features were available to train models. Three of these features were derived from the NCCTs, namely brain volume, intracranial cerebrospinal fluid (CSF) volume, and CSF ratio, due to their association with functional outcomes in thrombectomy patients.[15] The clinical data were pre-processed using a scikit-learn pipeline, which imputed missing features, consolidated certain categorical feature classes for simplicity, and standardized ordinal and continuous features to zero center the mean and scale to the unit variance. The imaging features included NCCT and CTA in the form of Digital Imaging and Communications in Medicine (DICOM) files, which were obtained from the regional hospital PACS. All images were de-identified with syngo.via (Siemens Healthcare) and further screened with DicomBrowser

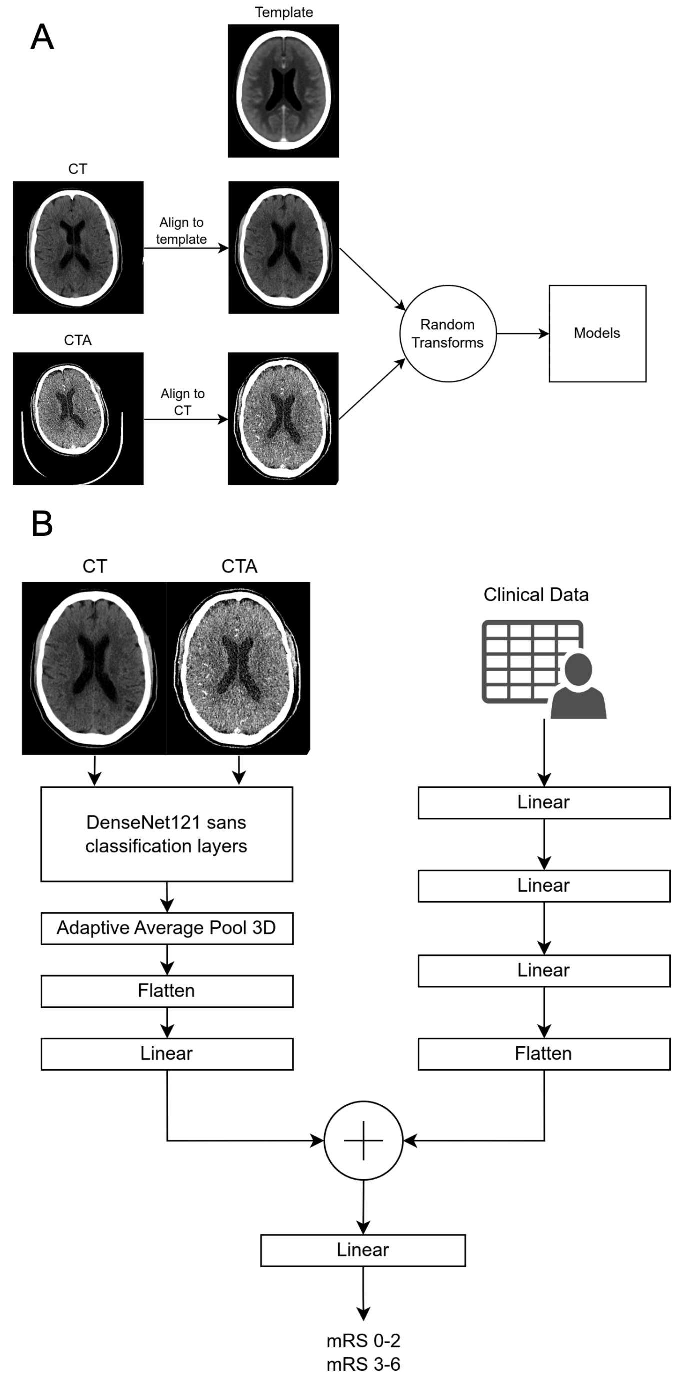

Figure 1 High-level overview of baseline computed tomography (CT) pre-processing steps (A) and high-level overview of the combined clinical and CT deep learning model architecture (B). CTA, computed tomography angiography; mRS, modified Rankin Scale.

(2010–2016 Neuroinformatics Research Group) and DicomCleaner (PixelMed Publishing™) to ensure full de-identification. Imaging pre-processing steps started by converting DICOM files to NiFTI format and resampling to a consistent voxel size. The CTs were further pre-processed (figure 1) with the Advanced Normalization Tools via the Extrantsr R package.[16] The NCCTs were registered to the Rorden *et al* CT template with a rigid translation and rotation.[17] The CTAs were cropped to isolate the head, removing the neck and torso, and were subsequently registered to the aligned NCCTs using a symmetric normalization transform, ensuring a seamless overlap between the two scan types through affine and deformation transformations. Data augmentations were applied to the CTs during training using MONAI data transformations. Image windowing was randomized to introduce random contrast changes. Other random transformations included selecting one of a set of random intensity transforms, random Gaussian smooth, random Gaussian sharpen, random Gaussian noise, and a random 3D affine.

Data partitions

The datasets were partitioned into training, testing, and validation partitions and were disjoint at the patient level. Data from patients in EVT center 1 were randomly split into training (70%) and testing sets (10%). Data from the EVT center 2 were used as the external validation set (20%). Due to the substantial amount of data required by deep learning models, we used a relatively small testing split of 10% to increase the amount of training data available to train our deep learning models.

Model development

We trained a set of classical machine learning models on clinical tabular data, deep learning models on CT images, and deep learning models on both clinical tabular data and CT images to predict functional independence (mRS 0–2). Models were developed using both anterior circulation and posterior circulation strokes, and anterior circulation strokes alone. The classical machine learning models included a Logistic Regression Classifier and a Random Forest Classifier. The deep learning models trained on CT images included a 3D DenseNet121,[18] with one input channel trained on the NCCTs and a 3D DenseNet121 with two input channels trained on both the NCCT and CTAs. The deep learning model trained on both clinical tabular data and CT images used a hybrid network architecture (figure 1) composed of an imaging network implemented with the DenseNet121 architecture (without its classification layer) to produce image embedding and a tabular data network composed of three fully connected layers. The embedding from the image network was concatenated with the last layer from the tabular network and fed into a fully connected layer for classification. The classical machine learning models were implemented with scikit-learn version 1.2.0,[19] and the deep learning models with MONAI (1.1.0)[20] and PyTorch.[21] Further details on model development and training hyperparameters are described in the online supplemental material. Finally, we evaluated the MR PREDICTS algorithm using their web-based tool on patients from EVT center 2.[5] The CTA collateral score was not included in our model development, but was calculated by one of the authors (KG) for patients in EVT center 2 to evaluate the MR PREDICTS algorithm.

Quality of dataset and deep learning pipeline and model evaluation

We assessed the quality of the dataset and the end-to-end deep learning pipeline by developing a parallel deep learning model that predicted patient sex based on NCCT, as this has previously been shown to successfully predict patient sex using NCCT images.[22] Statistical analysis was performed using R version 3.6.2 (R Foundation, Vienna, Austria) and MedCalc Statistical Software version 18.0 (Ostend, Belgium). The prognostic performance (C-statistics) of deep learning, the best performing classical machine learning model, and MR PREDICTS models were compared using the DeLong method. False discovery rate adjustment of p-values (Q-values) was applied to account for multiple comparisons. All tests were two-tailed, and $p<0.05$ was considered significant. Explainability methods were incorporated into the deep learning pipeline so that model decision-making for individual cases could be interrogated.[23] Gradient-weighted Class Activation Mapping (Grad-CAM) was used for CT model explainability. Local interpretable model-agnostic explanations (LIME) was used for clinical tabular model explainability. Grad-CAM uses the gradients of the target concept (ie, mRS 0–2 vs 3–6 in this instance) that feed into the final convolutional layer of a classification network to produce a heat map that highlights the most predictive regions in the image.[24] LIME are surrogate interpretable models that are used to explain individual predictions of machine learning models by showing the positive or negative relative impact of the model's most influential factors.[25]

RESULTS

There were 975 patients (550 men; mean±SD age 67.5±15.1 years) included in the study (table 1) with 778 patients in the model development cohort (EVT center 1) and 197 patients in the external validation cohort (EVT center 2). CT scans were performed with Philips (n=448 patients), Siemens (n=250), Toshiba (n=182), GE Medical Systems (n=94), and Canon Medical Systems (n=1) scanners. All patients were treated with EVT and 481 (49.3%) patients were also treated with intravenous thrombolysis. Successful recanalization was achieved in 848 (87.0%) patients. At 3 months, the median (IQR) mRS score was 2 (1–4), 521 (53.1%) patients achieved functional independence, and 147 (15.1%) patients had died. Compared with EVT center 1, EVT center 2 was associated with older age, female sex, European ethnicity, greater pre-stroke disability, and anterior circulation strokes; less dyslipidemia and smoking; higher systolic blood pressure, creatinine, and ASPECTS; lower hemoglobin; and a different profile of vessel occlusion. Features associated with poor clinical outcome are available in the online supplemental material.

The quality of the dataset and the end-to-end deep learning pipeline was high, as the deep learning model had an excellent classification accuracy for sex, with an area under the receiver operating curve (AUC) of 0.99 on the external validation cohort. The prognostic accuracy values for the anterior circulation deep learning, logistic regression, and MR PREDICTS models in predicting 3-month mRS scores are presented in table 2. The prognostic accuracy values for the combined anterior and posterior circulation deep learning and classical machine learning models (both logistic regression and random forests models) in predicting 3-month mRS scores are presented in the online supplemental material. Deep learning model 3 (trained on CT and clinical data) and the logistic regression model (clinical data alone) demonstrated the strongest discriminative abilities and were comparable (AUC 0.811 vs 0.817, Q=0.82). Both models exhibited superior prognostic performance to the other deep learning (CT alone, CTA alone) and MR PREDICTS models (all Q<0.05; online supplemental material). Figure 2 demonstrates examples of the explainability methods.

DISCUSSION

This study aimed to determine if deep learning on baseline CT images and clinical data improved prognostic accuracy in predicting functional independence when compared with classical prognostic methods in stroke patients treated with EVT. Our deep learning model had a high classification accuracy for sex, suggesting that the quality of the training, testing, and validation datasets was high, and that our end-to-end deep learning pipeline was implemented well.[26] Despite these factors, deep learning on baseline CT images and clinical data had a similar

Table 1 Demographic and clinical characteristics

Characteristic	All patients (N=975)
Centre, n (%)	
EVT center 1 (development cohort)	778 (79.8)
EVT center 2 (external validation cohort)	197 (20.2)
Demographics	
Age (years), mean±SD	67.5±15.1
Male sex, n (%)	550 (56.4)
European ethnicity, n (%)	639 (65.5)
Māori ethnicity, n (%)	141 (14.5)
Pacific ethnicity, n (%)	103 (10.6)
Asian ethnicity, n (%)	86 (8.8)
Other ethnicity, n (%)	6 (0.6)
Risk factors, n (%)	
Heart failure	173 (17.7)
Previous stroke	136 (13.9)
Diabetes	171 (17.6)
Hypertension	602 (61.7)
Dyslipidemia	402 (41.2)
Smoking	245 (25.1)
Coronary artery disease	191 (19.6)
Atrial fibrillation	472 (48.4)
Index stroke data	
Pre-stroke mRS, median (IQR)	0 (0–0)
Systolic blood pressure (mmHg), mean±SD	155±27
Glucose (mmol/L), median (IQR)	6.8 (6.0–8.3)
Hemoglobin (g/L), median (IQR)	139 (126–149)
Creatinine (µmol/L), median (IQR)	89 (74–102)
NIHSS, median (IQR)	16 (11–20)
Brain volume, median (IQR)	1.40 (1.29–1.50)
CSF volume, median (IQR)	0.13 (0.08–0.18)
CSF ratio, median (IQR)	0.09 (0.06–0.13)
ASPECTS, median (IQR)	9 (7–10)
Anterior circulation occlusion site, n (%)	873 (89.5)
IV thrombolysis, n (%)	481 (49.3)
Vessel, n (%)	
Anterior cerebral artery	1 (0.1)
Basilar artery	99 (10.2)
Internal carotid artery	224 (23.0)
M1 segment middle cerebral artery	505 (51.8)
M2 segment middle cerebral artery	143 (14.7)
Posterior cerebral artery	1 (0.1)
Vertebral	2 (0.2)
Side of occlusion, n (%)	
Bilateral	102 (10.5)
Left	466 (47.8)
Right	407 (41.7)
Procedural parameters	
LKN to puncture (min), median (IQR)	245 (180–400)
mTICI 2b-3 recanalization, n (%)	848 (87.0)
Outcome measures	
3-month mRS, median (IQR)	2 (1–4)
3-month functional independence, n (%)	521 (53.1)
3-month mortality, n (%)	147 (15.1)

ASPECTS, Alberta Stroke Program Early CT score; CSF, cerebrospinal fluid; EVT, endovascular thrombectomy; IQR, interquartile range; IV, intravenous; LKN, last known normal; mRS, modified Rankin Scale; mTICI, modified Thrombolysis in Cerebral Infarction; NIHSS, National Institutes of Health Stroke Scale; SD, standard deviation.

prognostic accuracy to logistic regression on clinical data alone. We speculate that this may be because there is a 'ceiling' in the ability of mathematical models to predict 3-month functional outcome without using intra-procedural or early post-procedural data.[27]

Many prognostic factors have been described in patients undergoing EVT, and a recent systematic review and meta-analysis identified 19 outcome prediction models for patients treated with EVT.[28] These 19 models were tested on a validation cohort of 3156 patients and reported discriminative performance ranging from an AUC of 0.61 (Stroke Prognostication Using Age and NIH Stroke Scale) to 0.80 (MR PREDICTS). The Totaled Health Risks in Vascular Events (THRIVE-c) and MR PREDICTS were the best calibrated models. Age, stroke severity, CT collateral score, and ASPECTS were the most widely used variables. MR PREDICTS was derived using logistic regression from a dataset of 500 patients,[5] and variables that predicted a functional independence included age, NIHSS score, pre-stroke mRS score, previous stroke, diabetes mellitus, systolic blood pressure, intravenous alteplase administration, ASPECTS, location of arterial occlusion, CTA collateral score, and onset to groin puncture time. However, the modest discriminative ability of the MR PREDICTS model has limited its widespread adoption in clinical decision-making.[28]

Other groups have attempted to use machine learning techniques to improve the discriminative ability of predictive models with varying success. A recent systematic review and meta-analysis of machine learning models identified 10 classical machine learning models and three deep learning models that predicted functional outcome in patients treated with EVT.[6] The pooled discriminative performance for classical machine learning and deep learning models were an AUC of 0.81 and 0.75, respectively. The machine learning models used between four and 53 features, with age, NIHSS, pre-stroke mRS, and ASPECTS being the most common variables.

A limitation of statistical and classical machine learning models is that they cannot use raw radiological data as an input. As a consequence, many potential 'hidden' markers of outcome are not utilized, for example, the degree of cerebral atrophy, which may act as a marker of 'brain reserve' and hence a good marker of functional recovery, is associated with functional outcome.[15] Intracranial carotid artery and vertebrobasilar artery calcification may also be markers of recovery capacity or of procedural difficulty.[29 30] The advantage of deep learning over statistical and classical machine learning models is that these or other obscure markers of outcome might be incorporated into the predictive powers of the model without a priori recognition.

In a systematic review and meta-analysis of 19 outcome prediction models for patients treated with EVT,[28] two reported deep learning models with data from CT and one with data

Table 2 Prognostic performance (area under the receiver operating curve) of various models in predicting modified Rankin Scale outcomes in the external validation cohort

Functional outcome model	Train AUC	Test AUC	Validation AUC (95% CI)
Deep learning model 1 (CT)	0.83 (0.79–0.87)	0.69 (0.57–0.81)	0.69 (0.62 to 0.76)
Deep learning model 2 (CT, CTA)	0.96 (0.94–0.98)	0.72 (0.60–0.84)	0.71 (0.64 to 0.77)
Deep learning model 3 (CT, CTA, clinical features)	0.80 (0.76–0.84)	0.76 (0.65–0.87)	0.81 (0.75 to 0.87)
Logistic regression model (clinical features)	0.76 (0.72–0.80)	0.72 (0.60–0.84)	0.82 (0.75 to 0.87)
MR PREDICTS model	NA	NA	0.74 (0.67 to 0.80)

AUC, area under the curve; CI, confidence interval; CT, computed tomography; CTA, CT angiography; NA, not available.

from MRI. The major pre-processing steps in these models using imaging data included skull stripping extraction, registration to a template, data augmentation, normalization, and image resampling. The model architectures included supervised 2D-ResNet with structured receptive field kernels model; image feature encoder, clinical metadata encoder, imaging, and clinical metadata fusion; a U-net segmentation task for imaging feature derivation, and a two-layer neural network for fine tuning. Only five of the 19 prediction models were validated externally, including only one deep learning model that segmented the ischemic lesion from 250 baseline MRI, and predicted the functional outcome of 74 patients with an AUC of 0.81.[31] The current study is the first to develop and externally validate a 3D convolutional neural network on baseline CT scans and clinical data to predict clinical outcome.

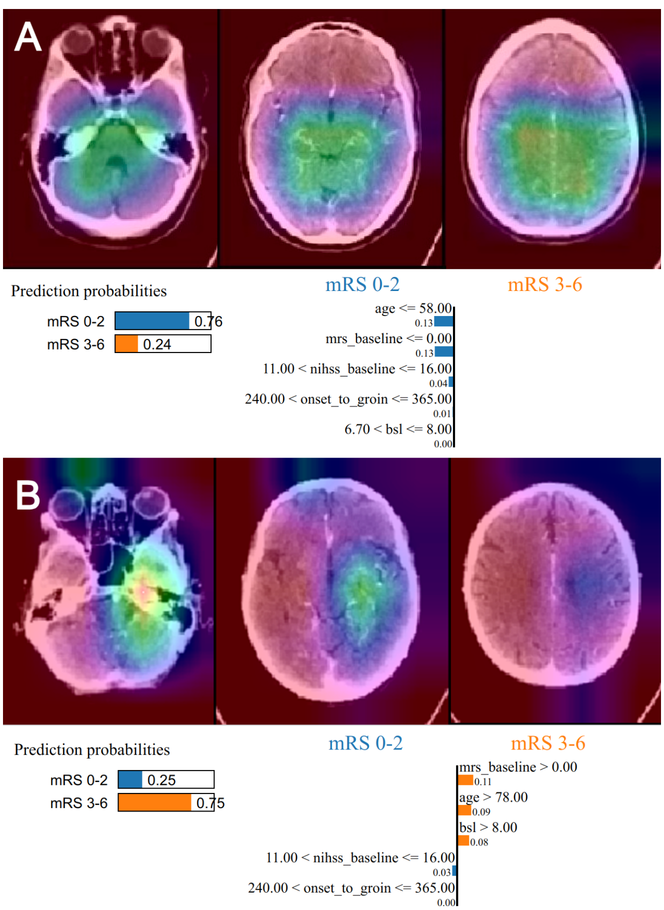

Figure 2 Deep learning explainability methods. Patient with left M1 occlusion, good collaterals, and a good clinical outcome (A) with Grad-CAM heat map centered towards the midline of CT angiogram. LIME visualizations showing age <58 years, mRS score ≤0, NIHSS between 11 and 16, and onset-to-groin time between 240 and 365 min, were all in favor of a good clinical outcome. Patient with a right M1 occlusion, poor collaterals, and a poor clinical outcome (B) with Grad-CAM heat map centered over the unaffected hemisphere of the CT angiogram. LIME visualizations showing mRS score >0, age >78 years, and blood sugar level >8 mmol/L were all in favor of a poor clinical outcome. NIHSS between 11 and 16 was in favor of a good clinical outcome. Grad-CAM, Gradient-weighted Class Activation Mapping; LIME, local interpretable model-agnostic explanations; mRS, modified Rankin Scale.

With the publication of the large ischemic core EVT studies there has been a shift away from pre-treatment prognostication as patients and families often prefer even a slim chance of neurological improvement.[32] However, we believe that prognostication still has utility for EVT patients, families, and healthcare services in the pre-procedural and immediate post-procedural period.[33] As an example, there is significant variation in EVT treatment practices between stroke patients younger or older than 80 years of age.[34] The decision as to whether to treat a frail elderly patient with early ischemic changes, significant pre-stroke disability, multiple medical comorbidities, and challenging procedural anatomy is not straightforward, particularly when they present to an outside hospital. In this situation, prognostic tools could assist with, and standardize, decision-making among stroke, neurointerventional, and neurocritical care teams, and provide referring physicians, patients, and families with realistic (and more standardized) outcome expectations before proceeding with treatment.

Informing patients and their family of the likely clinical outcome soon after treatment is challenging. Prognostic models may be of use to neurologists and rehabilitation physicians in the early post-procedural period, resulting in more standardized outcome predictions to guide their patients' management, including early rehabilitation and likely discharge disposition, and assisting families in planning long-term care.[33] The addition of Thrombolysis in Cerebral Infarction (TICI) reperfusion grade, NIHSS at 24 hours, and the presences of symptomatic intracranial hemorrhage at 24 hours has recently been shown to improve the prognostic performance of the MR PREDICTS tool with an AUC of 0.91.[27] The incorporation of these and other variables including post-procedural CT to our deep learning model may yield comparable or superior results and should be investigated in future studies.

This study has limitations. The model was developed from a small, single-center dataset with respect to machine learning practices. A larger development cohort might yield discriminative performance closer to 1 which would have greater clinical utility. The external validation cohort was also small and from the same country as the development cohort, limiting model generalizability. Missing data were imputed to avoid degradation of the model development cohort size which may have introduced bias. Neuroimaging was performed on several scanners using non-standardized imaging protocols which reflects everyday clinical practice. However, model development using standardized CT images might improve discriminative ability for a given dataset size. Although explainability methods were incorporated into the current model, they may need to be optimized to improve physician buy-in.[23]

The discriminative performance of our deep learning model for predicting functional independence was comparable to logistic regression models using clinical data and was not adequate for clinical utility. This may be because procedural and post-procedural factors are unknowable before EVT is performed, or because our development cohort size was too small. Future studies should focus on whether incorporating procedural and post-procedural data significantly improves model performance.

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Contributors WD conceived the concept of the research. WD, AM, GPT, KG, JB, and JH collected the data. JPD and T-YC developed the models. MTMW performed statistical analysis. TW, DC, and PAB provided oversight of the study. WD drafted the manuscript. All authors contributed to the writing and intellectual content of the article. JPD and WD contributed equally to the work. PAB is the guarantor of the study.

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Patient consent for publication Not applicable.

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Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request. The code for the final models are published open source on https://github.com/jdddog/deep-skull and https://github.com/jdddog/deep-mt. The pre-trained model weights and data that support the findings of this study are available from the corresponding author upon reasonable request.

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